

Exploration-of-AI-Use-Case-Case-study-on-self-driving-cars

AIM

To study the application of Artificial Intelligence (AI) in self-driving cars and understand how AI technologies help vehicles perceive their surroundings, make decisions, and navigate safely without continuous human control.

THEORY

Self-driving cars rely on a sophisticated pipeline of Artificial Intelligence (AI) systems to operate autonomously. The architecture is broadly categorized into three core modules: perception, decision-making (planning), and control.

Perception (Seeing and Sensing): Autonomous vehicles are equipped with an array of sensors, including cameras, LiDAR, radar, and ultrasonic sensors to gather real-time data about their environment. AI algorithms employ deep learning and computer vision to process this data. Through "sensor fusion," the vehicle merges data from multiple sources to accurately detect and track objects, pedestrians, traffic signs, and lane markings.

Decision-Making and Path Planning: Once the environment is mapped, machine learning models analyze the data to predict the behavior and trajectories of other road users. Path planning algorithms then calculate the safest and most efficient route. The AI assesses real-time risks to make complex driving decisions, such as determining safe following distances, changing lanes, and yielding right of way.

Motion Control: The decisions generated by the planning module are sent to the vehicle's motion control system. Specialized controller algorithms (such as Proportional Integral Derivative or PID controllers) translate these virtual decisions into physical actions, calculating the precise amount of steering, throttle, and braking required to smoothly execute the planned maneuver.

RESULT

The application of Artificial Intelligence in self-driving cars was successfully studied. It is evident that AI technologies—spanning sensor fusion, deep neural networks, and advanced control algorithms—are fundamental in enabling vehicles to accurately perceive complex, dynamic environments, make real-time intelligent decisions, and execute safe driving maneuvers without continuous human intervention.